

VisionFood QAI: 2-Week Model Training and Deployment Plan

Scope: packaged food products, cans, transparent bottles, label anomalies, and QR mismatch validation.

Timeline: 14 days | Team size: 3 people | Objective: edge-ready real-time deployment on same device.

1. Objectives and Deliverables

- Train and validate a production-style defect detection pipeline.
- Add label anomaly detection and QR mismatch verification.
- Integrate severity grading (S1-S4) and escalation logic.
- Deliver edge-exportable models (ONNX) and deployment checklist.

2. Data Sources (Collection Plan)

Source	Use in Project	Priority
MVTec AD	Primary anomaly patterns and baseline transfer learning	High
MVTec LOCO AD	Logical and structural anomaly robustness	Medium
VisA	Additional industrial anomaly texture diversity	Medium
KolektorSDD2 / BTAD	Hard-defect texture regularization	Low-Medium
In-house production data	Final domain adaptation for cans, bottles, labels, QR	Critical

Collection requirements:

- Collect positive and clean-negative samples from each SKU and container type.
- Store metadata per image: SKU, container type, camera ID, line shift, lighting condition.
- Include hard cases: glare, blur, reflections on transparent bottles, partial label occlusion.

3. Model Types and Training Method

Module	Model	Method
Defect Detection	YOLOv11n (fallback: YOLOv11s)	Transfer learning from COCO-pretrained weights; fine-tune on project classes
Label Anomaly Classification	EfficientViT-M2 or EfficientNet-B0	Train on label crops; focal loss + label smoothing; hard-case augmentation
QR Validation	QR region detector + decoder + rule checker	Detect QR region, decode text, compare against expected SKU/batch mapping
Severity Grading S1-S4	XGBoost (feature model)	Train on class + box area + confidence + uncertainty + retry count

Transfer learning recommendations:

- YOLOv11: initialize from COCO pretrained checkpoint and fine-tune full detector after warmup.
- Label classifier: initialize from ImageNet pretrained backbone and progressively unfreeze.

- Optional: pre-adapt anomaly branch with MVTec, then final fine-tune on in-house data only.

4. Two-Week Task Split Across 3 People

Roles

- Person A: Data Lead (collection, curation, annotation quality)
- Person B: ML Lead (model training, tuning, evaluation)
- Person C: Integration Lead (frontend + pipeline integration + deployment packaging)

Day	Person A - Data	Person B - ML	Person C - Frontend/Deploy
1	Freeze class taxonomy and source list	Define baseline config and experiment template	Define API-frontend contract and metrics view
2	Build manifest and split (train/val/test)	Prepare YOLO training pipeline	Create training dashboard placeholders
3	Annotation QA pass and fixes	Run YOLO baseline training	Hook live metrics endpoints in UI
4	Hard-case sample mining	YOLO tuned run with domain augmentations	Integrate inference result cards and error states
5	Finalize detector dataset v1	Detector threshold tuning and error analysis	Prepare model artifact registry screen
6	Prepare label crop dataset	Train label anomaly classifier v1	Connect label anomaly outputs to UI
7	Collect QR mismatch samples and labels	Implement and validate QR mismatch module	Frontend support for QR status and mismatch alerts
8	Severity label mapping audit	Train severity model (XGBoost) v1	Integrate severity S1-S4 in dashboard
9	Drift and class-balance review	Uncertainty calibration and thresholding	Add review/escalation UI workflows
10	Final data corrections for release set	End-to-end pipeline run on validation	Integrate full pipeline in app and API
11	Prepare edge benchmark set	Export ONNX and quantization tests	Latency instrumentation in frontend
12	Final QA on test subset	Benchmark accuracy plus latency on target device	Deployment scripts and config packaging
13	Compile evidence for report	Ablation and final model selection	Release-candidate smoke test
14	Documentation and handoff	Final metrics and model cards	Demo flow, deployment readiness checklist

5. Quality Gates and Targets

- Detector target: mAP50 $\geq$ 0.80, high-risk class recall $\geq$ 0.85.
- Severity target: macro F1 $\geq$ 0.70 on S1-S4.
- Latency target: preferred $\leq$ 120 ms per sample (acceptable $\leq$ 180 ms for capstone).
- Deployment target: ONNX export validated; rollback model retained.

6. Final Deliverables by End of Week 2

- Cleaned dataset manifest and annotation guideline.
- Best model artifacts: detector, label module, severity model, calibration config.
- Evaluation report: accuracy, confusion matrix, latency, failure cases.
- Frontend-integrated demo with PASS/FAIL/ESCALATE/REVIEW and severity display.
- Deployment package: ONNX models, threshold config, startup and monitoring checklist.